

Ai Rag Notes

Artificial Intelligence

Artificial Intelligence (AI) is the field of computer science concerned with building systems that can perform tasks associated with human intelligence, such as reasoning, learning, perception, and language understanding.

Machine Learning

Machine Learning is a subset of AI in which algorithms learn patterns from data and use those patterns to make predictions or decisions.

Generative AI

Generative AI systems create new content such as text, images, audio, video, and code. Large language models are commonly used for text generation.

Retrieval-Augmented Generation

RAG combines information retrieval with text generation. A RAG system retrieves relevant information from an external knowledge source and provides that information to a language model as context.

RAG Pipeline

A typical RAG pipeline loads documents, splits them into smaller chunks, creates embeddings for the chunks, stores the embeddings in a vector database, retrieves relevant chunks for a user query, and passes the retrieved context to a language model.

Embeddings

An embedding is a numerical representation of text. Embeddings allow a system to compare text based on semantic similarity rather than exact keyword matching.

Vector Database

A vector database stores embeddings and supports similarity search. It is commonly used in RAG systems to retrieve chunks that are semantically related to a query.